Course Title: Building Sciences and Mechanics
Course Code: 1BESC104A

Module-1 – Introduction to Building Science

Syllabus
Introduction to Civil Engineering:

- Surveying
- Structural Engineering
- Geotechnical Engineering
- Water Resources Engineering
- Transportation Engineering
- Environmental Engineering
- Construction Planning & Project Management.

Basic Materials of Construction:

- Types and Uses of Bricks
- Stones
- Cement
- Structural steel
- Wood
- Concrete

Structural elements of a building:

- Foundation
- Plinth
- Lintel
- Chejja
- Masonry Wall
- Column
- Beam
- Slab
- Flooring
- Staircase

Introduction to Civil Engineering

The scope of Civil Engineering is very vast, and it has many diversified fields which help in the total development of any civilization. Various subdivisions that come under civil engineering branch are listed below.

1. Surveying
2. Structural Engineering
3. Geotechnical Engineering
4. Hydraulics & Water Resources
5. Transportation Engineering
6. Environmental Engineering
7. Construction Planning & Project Management.

1. Surveying:

Surveying is a specialized field used to find the positions and heights of different points on the earth's surface. The measured details are shown on a map or plan, which represents the ground features to a suitable scale. Traditional methods of surveying include chain, compass, theodolite, plane table surveying, and leveling. With modern technology, advanced instruments like total station, GPS, GIS, and remote sensing are used to make surveying more accurate and faster

Importance of Surveying:

- Helps in planning and designing all civil engineering projects like buildings, roads, and bridges.
- Assists in preparing accurate maps and layouts of the land.
- Determines boundaries of properties to avoid land disputes.
- Helps in estimating quantities of earthwork, materials, and construction costs.

Scope of Surveying:

- Topographical Survey: Maps natural and man-made features of the land.
- Cadastral Survey: Defines property boundaries for legal purposes.
- Engineering Survey: Used in the design, construction, and maintenance of structures.
- Route and Hydrographic Survey: Helps in planning roads, railways, pipelines, and water projects.

2. Structural Engineering

Structural Engineering is a branch of Civil Engineering that deals with the design, analysis, and construction of structures such as buildings, bridges, towers, dams, and tunnels. It ensures that these structures can safely resist the loads and forces acting on them, such as weight, wind, earthquake, and temperature changes. The main aim of structural engineering is to create safe, stable, and durable structures using materials like concrete, steel, timber, and masonry.

Importance of Structural Engineering:

- Ensures the safety and stability of buildings, bridges, and other structures.
- Helps in efficient use of materials while maintaining strength and durability.
- Guides in the design of structures to withstand loads like wind, earthquake, and traffic.
- Supports the planning and execution of construction projects accurately.

Scope of Structural Engineering:

- Building Structures: Design and analysis of residential, commercial, and industrial buildings.
- Bridge Engineering: Planning and construction of safe and durable bridges.
- Special Structures: Design of towers, dams, stadiums, and high-rise buildings.
- Research and Development: Studying new materials, earthquake-resistant design, and innovative construction techniques.

3. Geotechnical Engineering

Geotechnical Engineering is a branch of Civil Engineering that deals with the study of soil and rock materials on which structures like buildings, bridges, dams, and roads are built. It helps engineers understand the behavior of the ground under different loading conditions. The main aim is to ensure that structures are safe, stable, and economical by designing suitable foundations based on soil properties.

Importance of Geotechnical Engineering:

- Helps in understanding soil and rock properties for safe foundation design.
- Ensures the stability of structures like buildings, bridges, and dams.
- Guides in preventing structural failures due to soil problems like settlement or landslides.
- Assists in efficient use of construction materials and cost-effective foundation solutions.

Scope of Geotechnical Engineering:

- Foundation Design: Designing shallow and deep foundations for different structures.
- Soil Investigation: Studying soil and rock conditions for construction projects.
- Slope Stability and Earth Retaining Structures: Designing embankments, retaining walls, and slopes safely.
- Ground Improvement Techniques: Improving weak soil using compaction, stabilization, or reinforcement methods

4. Water Resources Engineering

Water Resources Engineering is a branch of civil engineering that deals with the collection, storage, control, and distribution of water for various uses such as irrigation, domestic supply, industrial purposes, and hydroelectric power. It involves the planning, design, and management of systems like dams, canals, reservoirs, and pipelines to ensure the optimal use of water resources while minimizing wastage and environmental impact.

Importance of Water Resources Engineering:

- Ensures adequate supply of water for drinking, irrigation, and industries.
- Helps in flood control and drought management to protect lives and property.
- Supports hydropower generation and other energy needs.
- Promotes sustainable and efficient use of water resources.

Scope of Water Resources Engineering:

- Design of Dams and Reservoirs: For water storage and flood control.
- Irrigation Systems: Planning and managing canals, pipelines, and water distribution.
- Flood Management and Watershed Development: Preventing water-related disasters and conserving resources.
- Hydropower and Water Supply Projects: Planning and designing for energy and municipal water needs

5. Transportation Engineering

Transportation Engineering is a branch of civil engineering that deals with the planning, design, construction, and maintenance of transportation systems like roads, railways, airports, and urban transport. It ensures the safe, efficient, and economical movement of people and goods from one place to another.

Importance of Transportation Engineering:

- Ensures efficient and safe movement of people and goods by road, rail, air, and water.
- Supports economic development by connecting cities, industries, and markets.
- Helps in reducing travel time and transportation costs.
- Plays a key role in planning and maintaining infrastructure like roads, bridges, and airports.

Scope of Transportation Engineering:

- Highway Engineering: Planning, designing, and maintaining roads and pavements.
- Railway Engineering: Designing tracks, stations, and railway systems.
- Airport and Air Traffic Engineering: Planning and managing airports and air navigation.
- Urban Transport and Traffic Engineering: Designing city transport systems and managing traffic flow.

6. Environmental Engineering

Environmental Engineering is a branch of civil engineering that focuses on protecting and improving the environment. It deals with water and wastewater treatment, air pollution control, solid waste management, and sustainable use of natural resources to ensure health and safety for humans and the ecosystem.

Importance of Environmental Engineering:

- Ensures clean air, water, and soil for healthy living.
- Helps in controlling pollution from industries, cities, and agriculture.
- Supports sustainable development by conserving natural resources.
- Protects public health and the environment from harmful wastes and chemicals.

Scope of Environmental Engineering:

- Water and Wastewater Treatment: Designing systems to provide safe drinking water and treat sewage.
- Air Pollution Control: Reducing harmful emissions from industries and vehicles.
- Solid Waste Management: Collecting, processing, and disposing of municipal and industrial waste.
- Environmental Impact Assessment and Sustainability: Planning projects to minimize environmental damage and promote eco-friendly practices.

7. Construction Planning & Project Management

Construction Planning & Project Management is a branch of civil engineering that deals with organizing, scheduling, and controlling construction activities to complete projects efficiently. It involves planning resources, labor, materials, and time, as well as monitoring progress to ensure the project is completed on time, within budget, and as per quality standards.

Importance of Construction Planning & Project Management:

- Ensures projects are completed on time and within budget.
- Helps in efficient use of resources like labor, materials, and equipment.
- Improves coordination and communication among various teams on site.
- Minimizes risks, delays, and cost overruns during construction.

Scope of Construction Planning & Project Management:

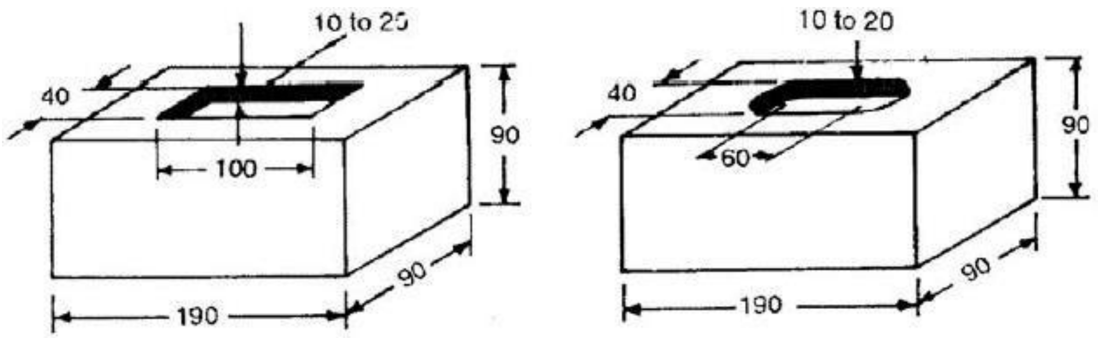
- Project Scheduling: Planning activities, timelines, and milestones for construction projects.
- Resource Management: Efficient allocation of labor, materials, and machinery.
- Cost Estimation and Control: Estimating budgets and controlling expenses.
- Risk Management and Quality Control: Identifying risks and ensuring construction quality standards are met.

BASIC MATERIALS OF CONSTRUCTION

BRICKS:

Brick is one of the oldest and most commonly used **building materials** in the world. It has been used since ancient times and continues to be popular due to its **low cost, durability, strength, and ease of handling**. Bricks are used in the construction of buildings, pavements, bridges, retaining walls, and other masonry works.

A brick is a **small, rectangular block** made typically from clay, sand, lime, or concrete, which is used as a **building unit in construction**. Bricks are joined together using a **binding material**, such as mortar, to form durable and stable structures like walls, pillars, and foundations.



Length of brick = 2 × width of brick + thickness of mortar
Height of brick = width of brick
(L: B: D = 4: 2:2)

Size of a standard brick (modular brick) should be 19 × 9 × 9 cm and 19 × 9 × 4 cm. When placed in masonry the **19 × 9 × 9 cm** brick with mortar becomes 20 × 10 × 10 cm. Weight of such a brick is **3.0 kg.** An **indent called frog**, 1–2 cm deep, is provided for 9 cm high bricks. The size of frog should be 10 × 4 × 1 cm. The purpose of providing frog is to form a **key for holding the mortar** (stamp the logo of manufacturer) and therefore, the bricks are laid with frogs on top.

CLASSIFICATION/TYPES OF BRICKS:

1) Based on Method of Manufacture
2) Based on Burning or Firing
3) Based on Quality
4) Based on Material Used

1. Based on Method of Manufacture

(a) Hand-Made Bricks:
These are manufactured manually by **moulding clay in hand moulds.** They are generally irregular in shape and size and used for ordinary construction work.

(b) Machine-Made Bricks:

These are manufactured using **machines,** ensuring uniform shape, size, and better quality. Machine-made bricks are stronger and more durable than hand-made ones.

2. Based on Burning or Firing

(a) Unburnt (Sun-Dried) Bricks:

These are dried naturally in the **sun and not burnt in kilns**. They are weak, less durable, and used for temporary or low-cost structures.

(b) Burnt Bricks:

These are **burnt in kilns** and are strong, durable, and used for permanent structures. Burnt bricks are further classified into four classes based on their quality.

3. Based on Quality (Burnt Clay Bricks)

(a) First-Class Bricks:

These are well-burnt, uniform in size, shape, and color, with sharp edges. They are free from cracks and flaws, have **high compressive strength,** and produce a clear ringing sound when struck together. Used for superior quality construction.

(b) Second-Class Bricks:

These are slightly **over-burnt or under-burnt,** with rough surfaces and irregular edges. Used for **internal walls or less important** structures.

(c) Third-Class Bricks:

These are **under-burnt,** soft, and of low strength. They are not suitable for load-bearing walls and are used for **temporary construction**.

(d) Fourth-Class Bricks:

These are **over-burnt and brittle,** often fused together in the kiln. They are not suitable for masonry but can be used as **aggregates in road construction or concrete.**

4. Based on Material Used

a) **Clay Bricks:** Made from natural clay and most commonly used.
b) **Fly Ash Bricks:** Made from fly ash, lime, and gypsum; smooth finish, high strength, and eco-friendly.
c) **Concrete Bricks:** Made from cement, sand, and aggregates; used in modern constructions.
d) **Fire Bricks (Refractory Bricks):** Made from special clay to withstand high temperatures; used in furnaces and chimneys.
e) **Sand-Lime Bricks:** Made from sand and lime; have good strength and smooth finish.

CHARACTERISTICS/QUALITIES OF GOOD BRICK

The essential requirements for building bricks are sufficient strength in crushing, regularity in size, a proper suction rate, and a pleasing appearance when exposed to view.

1. **Size and Shape:** The bricks should have **uniform size** and plane, rectangular surfaces with parallel sides and **sharp straight edges.**
2. **Colour:** The brick should have a uniform **deep red or cherry colour** as indicative of uniformity in chemical composition and thoroughness in the burning of the brick.
3. **Texture and Compactness:** The surfaces should **not be too smooth** to cause slipping of mortar. The brick should have **pre-compact** and uniform texture. A fractured surface should not show fissures, holes grits or lumps of lime.
4. **Hardness and Soundness:** The brick should be so hard that when scratched by a finger nail **no impression** is made. When two bricks are struck together, **a metallic sound** should be produced.
5. **Water absorption** should **not exceed 20%** of its dry weight when kept immersed in water for 24 hours.
6. **Crushing strength** should not be less than **10 N / mm^2**.
7. **Brick earth** should be free from stones, kankars, organic matter etc.

RAW MATERIALS or INGREDIENTS OF BRICKS:

1) Alumina or Clay:

- 20 to 30%
- Absorbs water and imparts plasticity
- Thus bricks can be moulded to desired shape
- Excess contents will result in shrinkage

2) Silica or Sand:

- 50 to 60%
- Prevents cracking and shrinkage of raw materials
- Excess contents makes the bricks brittle

3) Lime (CaO):

- Not exceeding 5%
- Sand alone is not fusible but in presence of lime it fuses
- Excess of lime makes bricks to melt

4) Iron Oxide (Fe_2O_3):

- 5 to 6% is desirable
- Imparts colour to bricks
- Gives strength and hardness
- Excess content makes bricks dark blue or blackish

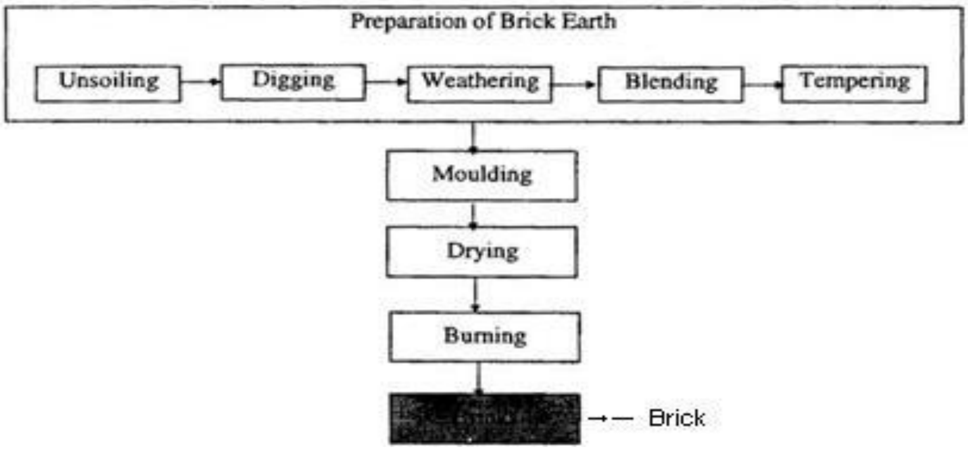
5) Magnesia (MgO):

- 1 to 2%
- Imparts yellow tint to bricks
- Decreases shrinkage
- Excess content leads to decay of bricks

MANUFACTURING PROCESS OF BRICKS:

There are 4 different operations are involved in the process of manufacturing of bricks:

1. **Preparation of clay**
2. **Moulding**
3. **Drying**
4. **Burning**



1. Preparation of clay for brick manufacturing:

- **Unsoiling:** The top layer of soil, about **20 cm thick**, is removed because it contains grass, roots, and other impurities. Only the clean clay below this layer is used for making bricks.
- **Digging:** After removing the topsoil, the **clay is dug out** from the ground and spread on a level surface. This helps it to dry and makes it easier to work with.
- **Cleaning:** The dug-out clay is cleaned to **remove** stones, pebbles, and vegetable matter. If many impurities are present, the clay is washed and screened. The lumps are crushed to make the clay fine and uniform.
- **Weathering:** The cleaned clay is left open to the atmosphere for **3 to 4** weeks or during a rainy season. This natural exposure helps the clay to soften, making it easier to mould later.
- **Blending:** At this stage, materials like **sand or fly ash** are added to the clay to improve its quality. The ingredients are mixed well by turning and kneading the clay until it becomes uniform in texture.
- **Tempering:** Finally, water is added to the clay, and it is thoroughly mixed to make it **soft and plastic.** This is done by the feet of men or animals for small work, or by using a pug mill for large-scale brick making. The clay is now ready for moulding into bricks.

2. Moulding of clay for brick manufacturing

In the molding process, prepared clay is mould into brick shape (generally rectangular). This process can be done in two ways according to scale of project.

1. Hand molding of bricks

This method is commonly used when bricks are made on a small scale or by hand. It can be done in two types:

a) Ground Moulding:

- The ground is leveled and sprinkled with sand to prevent the clay from sticking.
- The mould is also dipped in water or sanded.
- The moulder fills the clay into the mould and levels it with a strike.
- The mould is then lifted, and the brick is left on the ground to dry.

b) Table Moulding:

- This method is similar to ground moulding but done **on a table** instead of the ground.
- It is faster and cleaner, often used when better-quality bricks are needed.

2. Machine molding of bricks

This method is used when bricks are produced on a large scale. Machines help to shape bricks more quickly and uniformly.

- The clay is placed in a machine called a brick moulding machine.
- The machine presses or extrudes the clay into moulds, forming bricks of uniform size and smooth finish.
- This method requires less labor and produces stronger, denser bricks.

3. Drying of raw bricks

The green bricks are **arranged in rows** on a clean, level ground under shade or in drying sheds. There should be enough space between the bricks for free **circulation of air.** Bricks are usually dried by **natural air and sunlight**. This process may take **7 to 14 days**, depending on the weather. The bricks are **turned at intervals** to ensure uniform drying on all sides. In large-scale production, artificial drying is used. **Hot air from kilns or furnaces** is passed over the bricks to speed up the drying process, especially in rainy or humid seasons.

4. Burning of bricks

In the process of burning, the dried bricks are burned either in **clamps** (small scale) **or kilns** (large scale) up to certain degree temperature. In this stage, the bricks will **gain hardness and strength**. The temperature required for burning is about **1100°C.** If they burnt **beyond this limit** they will be **brittle** and easy to break. If they burnt **under this limit**, they will **not gain full strength** and there is a chance to absorb moisture from the atmosphere.

TESTS ON BRICKS

Field Tests on Brick:

1) Shape and Size of Clay Bricks

The clay bricks should have a **uniform rectangular plan surface**, as per standard size and sharp straight edges. BSI recommends the standard size of brick is **190 mm x 90 mm x 90 mm** and constructional size is 200 mm x 100 mm x 100 mm.

2) Visual inspection

The bricks of good quality should be uniform in shape and should have truly **rectangular shape with sharp edges.**

3) Hardness of Clay Bricks

The clay bricks should be sufficiently hard when scratched by a **finger-nail no impression** should be left on the brick surface.

4) Colour of Clay Bricks

The clay bricks should have a uniform **deep red colour** throughout. It indicates the uniformity of chemical composition and the quality of burning of the bricks.

5) Texture and compactness of Clay Bricks

The surfaces should **not be so smooth** to cause skidding of mortar. The clay brick should have a **pre-compact,** homogeneous and uniform texture. A broken surface should be free form **cracks, holes grits or lumps of lime.**

6) Soundness of Clay Bricks

When 2 clay bricks are stuck together, **a metallic ringing** sound should come.

7) Structure

A brick is broken and its structure is examined. It should be homogeneous, compact and free from defects such as holes, lumps etc.

8) Basic Strength of Clay Bricks

When dropped flat on the hard ground from a height of about 1m, clay bricks should not break.

Laboratory Tests on Brick:

1) Water Absorption of Bricks

Water absorption of bricks should **not more than 20%** by its dry weight.

2) Compressive Strength of Brick

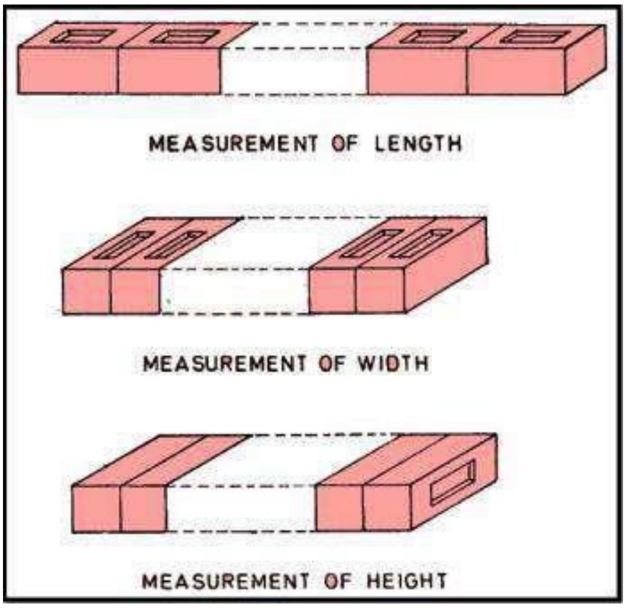
The brick is immersed in **water for 24 hours** and then placed in a **compression testing machine (CTM).** The load is applied gradually until the brick fails. The compressive strength is calculated by dividing the maximum load by the area of the brick. For first-class bricks, the minimum strength should be **10.5 N/mm^2.**

3) Efflorescence.

To check the presence of **soluble salts** in bricks, which cause white patches on the surface. Place the brick in a dish containing water (up to 25 mm depth). Allow the water to evaporate and observe the surface**. If white patches appear, efflorescence is present.** The brick is graded as nil, slight, moderate, heavy, or serious based on the extent of patches.

4) Dimension tolerance

Twenty bricks are selected at random to **check measurement** of length, width and height. The size variation should not exceed **±3 mm in length and ±1.5 mm in width or height.**



STONES

Stone is a naturally occurring solid material formed from one or more minerals. It has been used as a building material since ancient times for construction of foundations, walls, bridges, dams, monuments, etc. Stones are obtained from quarries and are used in dressed or undressed form depending on the requirement.

Classification of Stones

Stones can be classified based on:

1. **Geological Classification**

Igneous Rocks – Formed by the cooling and solidification of molten magma.
Examples: Granite, Basalt, Trap.
Uses: Foundations, road metal, bridges, etc.

Sedimentary Rocks – Formed by deposition and compression of sediments in layers.
Examples: Limestone, Sandstone, Shale.
Uses: Building blocks, paving stones.

Metamorphic Rocks – Formed from igneous or sedimentary rocks under heat and pressure.
Examples: Marble (from limestone), Slate (from shale), Gneiss (from granite).
Uses: Flooring, decorative works, roofing.

2. **Physical Classification**

Stratified Stones: Have layers; can be split easily (e.g., Sandstone, Limestone).

Unstratified Stones: Do not show layers (e.g., Granite, Trap).

Foliated Stones: Have a flaky structure; can be split along planes (e.g., Gneiss).

3. **Chemical Classification**

Siliceous Stones: Mainly silica; hard and durable (e.g., Granite, Quartzite).
Argillaceous Stones: Mainly alumina; soft and durable if well-burnt (e.g., Slate, Laterite).
Calcareous Stones: Mainly calcium carbonate (e.g., Limestone, Marble).

Requirements / Characteristics of Good building stones

- **Strength:** Should be strong enough to bear loads and stresses.
- **Durability:** Should resist weathering and remain intact for long periods.
- **Hardness:** Should resist wear and abrasion effectively.
- **Toughness:** Should withstand impact and vibrations without breaking.
- **Specific Gravity:** Should have a value between 2.4 and 2.8 for good quality.
- **Porosity:** Should have low porosity and absorb less than 1% water by weight.
- **Appearance:** Should have a uniform color and pleasing texture.
- **Workability:** Should be easy to cut, dress, and shape as required.
- **Resistance to Fire:** Should withstand high temperatures without cracking.
- **Resistance to Chemicals:** Should resist acids, alkalis, and corrosive gases.
- **Soundness:** Should produce a clear ringing sound when struck with a hammer.
- **Economy:** Should be easily available and economical in cost.

Uses of Stones in Civil Engineering works

- **Building Construction:** Used for walls, foundations, pillars, and arches.
- **Flooring and Paving:** Used as slabs for floors, pavements, and courtyards.
- **Road Works:** Crushed stones used as road metal, ballast, and aggregates in concrete.
- **Bridges and Dams:** Used for masonry work, retaining walls, and river training structures.
- **Roofing and Cladding:** Thin stones like slate used for roofing and wall facings.
- **Monuments and Decorative Works:** Used for statues, memorials, and ornamental facades.
- **Retaining and Boundary Walls:** Used to resist earth pressure and mark property limits.

- **Water Structures:** Used in lining of canals, reservoirs, and embankments.
- **Cement Manufacture:** Limestone used as a raw material for cement production.
- **Drainage Works:** Used in stone pitching and filter layers for water flow control.

Tests on stones

To ensure the suitability of stones for construction, several tests are performed:

1. **Crushing Strength Test:** Determines the compressive strength of stone; strong stones should have a strength of 100 N/mm^2 or more.
2. **Water Absorption Test:** The stone specimen is soaked in water for 24 hours; good stones absorb less than 1% of water by weight.
3. **Hardness Test:** Measured by scratching with a knife or by a Dory's testing machine; assesses resistance to wear.
4. **Smith's Soundness Test:** Two stones are struck together; a clear ringing sound indicates good quality and density.
5. **Acid Test:** Stone is exposed to dilute hydrochloric acid; formation of bubbles or powder indicates the presence of calcium carbonate and poor durability in acidic environments.
6. **Impact Test:** Used for stones in road works; assesses resistance to sudden shocks and vibrations.

Quarrying and Dressing of Stones

A. Quarrying of Stones

Quarrying is the process of extracting stones from natural rock beds or quarries.
Methods of Quarrying

- **Blasting Method:**

Holes are drilled in the rock and filled with explosives to break large blocks.
Suitable for hard rocks like granite and basalt.

- **Wedging Method:**

Holes are made and iron wedges are inserted and hammered to split the rock.
Used for medium-hard stones.

- **Channeling or Wire Saw Method**:

Diamond wire saws or machines cut blocks with precision.
Suitable for soft stones like marble and limestone.

B. Dressing of Stones

Dressing means shaping and finishing the quarried stones to make them suitable for use.

Types of Stone Dressing

- **Rough Dressing:** Removing irregularities and unwanted projections.
- **Hammer Dressing:** Making the surface roughly plane with hammer blows.
- **Fine Dressing:** Achieving a smooth surface using chisels and other tools.
- **Polished Dressing:** Producing a glossy, smooth finish for decorative use.

CEMENT:

Cement is a **binder,** a substance used for construction that sets, hardens, and adheres to other materials to bind them together. Cement is seldom (rarely) used on its own, but rather to bind fine aggregate (sand) and coarse aggregate (gravel) together. Cement mixed with fine aggregate produces mortar for masonry, or with fine aggregate and coarse aggregate, produces concrete.

HYDRAULIC CEMENT:

Hydraulic cements, such as Portland cement, have the special property of **setting and hardening** when mixed with water. This happens due to a chemical reaction between the dry cement ingredients and water, known as **hydration.** During this reaction, mineral hydrates are formed. These compounds are not soluble in water, which makes the cement **strong, durable, and water-resistant.** Because of this property, hydraulic cement can set and harden even in wet conditions or under water. The chemical process for hydraulic cement was found by ancient Romans who used volcanic ash (pozzolana) with added lime (calcium oxide).

BOGUES COMPOUNDS

When **water is added** to cement, it react with the ingredients of the cement **chemically and results** in the formation of complex chemical compounds terms as BOGUES compounds.

1. **C_3A -** Tri-Calcium Aluminate ($3CaO.Al_2O_3$) --------------------------8-12%
2. **C_4AF -** Tetra Calcium Alumino Ferrate ($4CaO.Al_2O_3.Fe_2O_3$) ----------- 6-10%
3. **C_3S -** Tri-Calcium Silicate ($3CaO.SiO_2$) -----------------------------30-50%
4. **C_2S -** Di-Calcium Silicate ($2CaO.SiO_2$) ------------------------------- 20-45%

1. Tri-Calcium Aluminate ($3CaO.Al_2O_3$ or C_3A)

- Formed in 24 hrs of addition of water
- Maximum evolution of heat of hydration
- Check setting time of cement

2. Tetra Calcium Alumino Ferrate ($4CaO.Al_2O_3.Fe_2O_3$ or C_4AF)

- Formed within 24 hrs of addition of water
- High heat of hydration in initial periods
- Contributes to the color of cement and helps in fusion during clinker formation. Has little effect on strength.

3. Tri-Calcium Silicate ($3CaO.SiO_2$ or C_3S)

- Formed within week
- Responsible for initial strength of cement
- Contribute about 50-60% of strength

4. Di-Calcium Silicate ($2CaO.SiO_2$ or C_2S)

- Last compound formed during hydration of cement
- Responsible for progressive later stage strength
- Structure requires later stages strength proportion of this component increase e.g. hydraulic structures, bridges.

Types of Cement

1. Ordinary Portland Cement (OPC)
2. Portland Pozzolana Cement (PPC)
3. Rapid Hardening Cement
4. Quick setting cement
5. Low Heat Cement
6. Sulfates resisting cement
7. Blast Furnace Slag Cement
8. High Alumina Cement
9. White Cement
10. Colored cement

CHEMICAL COMPOSITION OF OPC

COMPOUNDS	PERCENTAGE
Lime (CaO)	60 to 67%
Silica (SiO_2)	17 to 25%
Alumina (Al_2O_3)	3 to 8%
Iron oxide (Fe_2O_3)	0.5 to 6%
Magnesia (MgO)	0.1 to 4%
Sulphur trioxide (SO_3)	1 to 3%
Soda and/or Potash (Na_2O+K_2O)	0.5 to 1.3%

PHYSICAL PROPERTIES OF OPC

Properties	Values
Specific Gravity	3.16
Normal Consistency	29%
Initial Setting time	65min
Final Setting time	275 min
Fineness	330 kg/m^2
Soundness	2.5mm
Bulk Density	830-1650 kg/m^3

Testing of cements:

❖ ***Field Tests of Cement***

1. Colour Test of Cement

The colour of the cement should **not be uneven.** It should be a uniform grey colour with a **light greenish shade**.

2. **Presence of Lumps**

   The cement should not contain any **hard lumps.** These lumps are formed by the absorption of moisture content from the atmosphere. The cement bags with lumps should be avoided in construction.
3. **Cement Adulteration Test**

   The cement should be smooth if you **rubbed it between fingers.** If not, then it is because of adulteration with sand.
4. **Float Test**

   The particles of cement should **flow freely in water** for some time before it sinks.
5. **Date of Manufacturing**

   It is very important to check the manufacturing date because the strength of cement **decreases with time**. It's better to use cement **before 3 months** from the date of manufacturing.

❖ *Laboratory Tests of Cement:*

1. **Fineness Test**

- Determines how fine the cement particles are.
- Finer cement gives more strength and faster setting.
- Done using a **90-micron sieve** or Blaine's air permeability apparatus.

2. **Consistency Test**

- Finds the amount of water needed to prepare a cement paste of standard consistency.
- Performed using **Vicat apparatus**.

3. **Setting Time Test**

- Measures **initial** and **final setting times** of cement.
- Ensures cement does not set too quickly or too slowly.
- Conducted using **Vicat apparatus**.

4. **Soundness Test**

- Checks the ability of cement to resist expansion after setting.
- Prevents cracks and disintegration.
- Tested using **Le Chatelier apparatus**.

5. **Compressive Strength Test**

- Determines the **crushing strength** of cement mortar cubes.
- Conducted using a **compression testing machine**.

6. **Specific Gravity Test**

- Finds the density of cement compared to water.
- Helps in mix design calculations.
- Done using **Le Chatelier flask** or **pycnometer**.

7. **Heat of Hydration Test**

1. Measures the heat released during the reaction of cement with water.
2. Important for **mass concrete works**.
3. Done using a **calorimeter**.

CONCRETE

Concrete is a mixture of cement, fine aggregate (sand), coarse aggregate (gravel or crushed stone), and water. It is one of the most widely used construction materials in civil and structural engineering. It can be cast into any desired shape, and its strength increases with age.

Ingredients of Concrete

1. **Cement:**
    - Acts as a binding material.
    - Commonly used type: *Ordinary Portland Cement (OPC)*.
2. **Fine Aggregate (Sand):**
    - Fills voids between coarse aggregates.
    - Should be clean, well-graded, and free from silt or organic matter.
3. **Coarse Aggregate:**
    - Provides strength and reduces shrinkage.
    - Usually 20 mm or 40 mm size crushed stone or gravel.
4. **Water:**
    - Required for hydration of cement and workability.
    - Should be clean and free from acids, oils, or salts.
5. **Admixtures (Optional):**
    - Chemicals added to modify properties (e.g., plasticizers, retarders, accelerators).

Types of Concrete

1. **Plain Cement Concrete (PCC):**
    - Mixture of cement, sand, and aggregates without reinforcement.
    - Used in floors, pavements, and foundations.
2. **Reinforced Cement Concrete (RCC):**
    - Concrete reinforced with steel bars to resist tension.
    - Used in beams, slabs, columns, and bridges.
3. **Pre-stressed Concrete:**
    - Steel is tensioned before concrete is cast to improve strength.
    - Used in bridges, tanks, and long-span structures.
4. **Lightweight Concrete:**
    - Made with light aggregates like pumice or expanded clay.
    - Used in precast panels and thermal insulation.
5. **High-Density Concrete:**
    - Made with heavy aggregates like barytes; used for radiation shielding.
6. **Ready-Mix Concrete (RMC):**
    - Prepared in a batching plant and transported to site for use.

Properties of Good Concrete

- **Workability:** Concrete should be easy to mix, transport, place, and compact without segregation or bleeding.
- **Strength:** It should have adequate compressive and tensile strength to safely bear the applied loads.
- **Durability:** Good concrete should withstand weathering, chemical attack, and abrasion for a long service life.
- **Impermeability:** It should be dense enough to resist the entry of water and harmful chemicals.
- **Cohesiveness:** The mix should remain uniform, preventing separation of ingredients during handling.
- **Shrinkage:** It should have low shrinkage to minimize the formation of cracks during drying.
- **Resistance to Weathering:** It should resist alternate cycles of wetting, drying, freezing, and thawing.
- **Economy:** The concrete should be cost-effective while maintaining the required quality and strength.
- **Finish and Appearance:** The surface should be smooth, free from honeycombs, and pleasing in appearance.
- **Fire Resistance:** It should resist high temperatures without losing strength or integrity.

The stages of concrete production are:

- Batching or measurement of materials
- Mixing
- Transporting
- Placing
- Compacting
- Curing
- Finishing

STEEL

Steel is an alloy of iron and carbon containing less than 2% carbon, 1% manganese, and small quantities of silicon, phosphorus, sulphur, and oxygen. It is the most important engineering and construction material, widely used in buildings, vehicles, machinery, ships, and tools. Steel is 100% recyclable and can be reused repeatedly without loss of properties.

Properties of steel

1. HARDNESS It is the ability of steel to resist wear, abrasion, and indentation. Hardness is distinct from strength or toughness and is important in tools and cutting applications.

2. TOUGHNESS Toughness is the ability of a material to absorb energy without fracturing. A tough material resists cracks and breaks even when deformed under stress

3. YEILD strength Yield strength is the stress at which steel begins to deform permanently. It indicates the limit of elastic behavior beyond which plastic deformation starts.

4. TENSILE strength is a measurement of the force required to break the material.

5. ELONGATION (or Ductility) It is the ability of steel to stretch or deform before breaking. Expressed as a percentage, it shows how flexible or ductile the material is.

6. FATIGUE strength is the highest stress that a material can withstand for a given number of cycles without breaking.

7. CORROSION Corrosion is the chemical or electrochemical deterioration of steel due to exposure to moisture, oxygen, or acids. Protective coatings or alloying (e.g., stainless steel) help resist corrosion.

8. PLASTICITY is the deformation of a material undergoing non-reversible changes of shape in response to applied forces.

9. MALLEABILITY Malleability is the property that allows steel to be hammered or rolled into thin sheets without cracking. It indicates the ability to deform under compression

10. CREEP Creep is the slow, time-dependent deformation of steel under constant stress at high temperatures. It is significant in boilers, turbines, and high-temperature components.

TESTS ON STEEL REBAR

1. Tensile test
2. Compression test
3. Bending test
4. Brinell hardness test
5. Rockwell hardness test
6. Impact test
7. Torsion test

Uses of steel:

- Steel is used in reinforced concrete to provide strength and flexibility to structures.
- It is used in making beams, columns, and trusses for buildings and bridges.
- Steel bars and meshes are used in slabs, foundations, and retaining walls.

- Structural steel is used in high-rise buildings and industrial sheds.
- It is used in the construction of railway tracks and foot over-bridges.
- Steel pipes are used for water supply, drainage, and sewage systems.
- It is used in making gates, grills, and doors for buildings.
- Steel is used in the fabrication of formwork and scaffolding.
- It is used in construction machinery and equipment like cranes and mixers.
- Stainless steel is used in architectural works and for aesthetic finishes.
- Steel is environment-friendly & sustainable. It possesses great durability.
- Compared to other materials, steel requires a low amount of energy to produce lightweight steel construction.
- Steel is the world's most recycled material which can be recycled very easily. Its unique magnetic properties make it an easy material to recover from stream to be recycled.
- Mild steel is used for building construction. It is also a highly favored building frame material.

WOOD

Wood is one of the oldest and most widely used materials in engineering and construction. It is an **organic material** obtained from trees and is composed mainly of **cellulose fibers** (which give strength) and **lignin** (which binds the fibers together). Wood has several desirable properties such as **strength, toughness, workability, low weight**, and **aesthetic appeal**. It is used in **buildings, furniture, bridges, formwork, flooring, scaffolding, and carpentry works**.

Structure of Wood

A tree trunk consists of several concentric layers, each with specific functions and properties:

- **Bark:**

The outermost protective layer that shields the inner wood from mechanical injury, insects, and weathering. It consists of an **outer dead layer** and an **inner living layer** that helps in the transport of food.

- **Cambium Layer:**

A thin layer of living cells found just below the bark. It is responsible for the **growth of wood in thickness** by forming new sapwood cells every year.

- **Sapwood:**

The outer light-colored portion of the wood. It is **younger, softer**, and **contains sap** (water and dissolved nutrients). Although less durable, it plays an important role in the **transport of water** from roots to leaves.

- **Heartwood:**

The inner dark-colored, older portion of the tree trunk. It is **hard, dense, and more durable**, as the sap in it is replaced by natural oils and resins. Heartwood is the **most suitable part for engineering use**.

- **Pith (Medulla):**

The central core, soft and weak in older trees. It represents the **first year's growth** and is often removed in timber processing.

- **Annual Rings:**

The concentric circles visible on a cross-section of a tree trunk. Each ring represents **one year's growth** of the tree and helps determine its age and quality.

3. Classification of Wood

(a) Based on Botanical Source

Softwood:

- Obtained from **coniferous (needle-leaved) trees** like pine, cedar, and fir.
- Contains resin ducts and is generally light in color and weight.
- Has **straight grains**, making it easy to work.
- Used for temporary and lightweight structures like **doors, window frames, and packing cases**.

Hardwood:

- Obtained from **broad-leaved deciduous trees** like teak, oak, mahogany, and sal.
- Heavier, denser, and stronger than softwood.
- More durable and resistant to wear and moisture.
- Commonly used for **furniture, flooring, and heavy structural works**.

(b) Based on Suitability for Engineering Use

- **Timber:** Wood that is suitable for building and structural purposes.
- **Lumber:** Timber that has been sawn into planks, beams, or boards.

4. Characteristics of Good Timber

- **Strength:** Should be able to resist bending, compression, and tensile stresses.
- **Durability:** Should resist decay, moisture, and attack by insects and fungi.
- **Hardness:** Should be capable of resisting wear and abrasion, especially for floors and tool handles.
- **Toughness:** Should resist shocks and vibrations, important for tool handles, wagon parts, etc.
- **Appearance:** Should have a uniform color and pleasing texture without knots or stains.
- **Workability:** Should be easy to saw, plane, and nail without splitting.
- **Defect-free:** Should be free from knots, shakes, cracks, or resin pockets.
- **Well-seasoned:** Should be properly dried to reduce shrinkage and warping.

5. Seasoning of Wood

Seasoning is the process of **removing moisture from freshly cut wood** to improve its quality and durability. Freshly cut wood contains **30–40% moisture**, which must be reduced to about **10–15%** for use.

Objectives of Seasoning:

- To increase strength, hardness, and durability.
- To reduce weight for easier handling.

- To minimize warping, cracking, and shrinkage.
- To make wood resistant to decay and insect attack.

Methods of Seasoning:

1. **Natural (Air) Seasoning:**
    - Wood logs are stacked under a shed with proper air circulation.
    - It is simple and economical but slow (takes 6–12 months).
    - Commonly used in rural areas.
2. **Artificial (Kiln) Seasoning:**
    - Wood is placed in a **controlled chamber (kiln)** with regulated heat, humidity, and air flow.
    - It is faster (takes a few days) and results in **uniform, high-quality timber**.
    - Used for commercial and industrial purposes.

6. Defects in Timber

Defects are imperfections that reduce the strength, appearance, and durability of wood.

- **Knots:** Dark circular marks formed where branches were attached. Reduce strength and make machining difficult.
- **Shakes:** Cracks between annual rings caused by shrinkage. Types: **star shakes, heart shakes, ring shakes**.
- **Warping:** Distortion or twisting of wood due to uneven drying. Leads to uneven surfaces.
- **Splits and Checks:** Long narrow cracks along the grain due to rapid drying. Reduce strength and allow moisture entry.
- **Decay:** Caused by fungi or insects (termites, beetles) in the presence of moisture. Makes wood soft and discolored.

7. Preservation of Timber

Preservation protects wood from biological and environmental damage.

Objectives:

- To prevent attack by fungi, insects, and marine organisms.
- To increase life and durability.
- To make timber suitable for outdoor and underwater use.

Common Preservatives:

1. **Creosote Oil:**

   Oily liquid derived from coal tar; repels insects and moisture.

   Commonly used for railway sleepers, poles, and piles.

2. **Coal Tar or Paint:**

   Forms a waterproof and weather-resistant coating.

   Suitable for exposed surfaces.

3. **Chemical Salts:**

   Solutions like **copper sulphate** and **zinc chloride** are used for anti-termite protection.

Methods of Application:

- **Brushing or Spraying:** Coating the surface with preservative.

- **Soaking or Dipping:** Immersing timber in preservative solution.
- **Pressure Treatment:** Forcing preservative deep into wood under pressure (best method).

8. Uses of Wood in Engineering Works

- **Building Construction:** For doors, windows, beams, rafters, and roofs.
- **Furniture and Fixtures:** Tables, chairs, wardrobes, and cabinets.
- **Formwork and Scaffolding:** Temporary supports during concrete work.
- **Bridges and Marine Works:** Timber piles, planks, and fenders.
- **Railway Sleepers:** For supporting railway tracks.
- **Carpentry and Packaging:** Used in tool handles, boxes, and crates.

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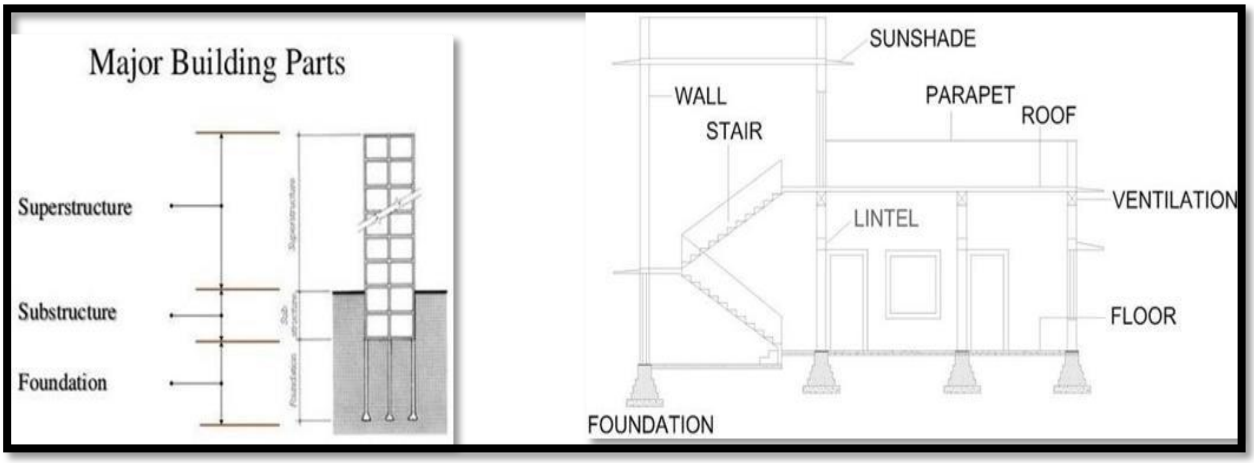
STRUCTURAL ELEMENTS OF A BUILDING

Building is an assemblage of various structural components like beams, columns, slabs and footings which resists deformation caused due to external loads. Therefore buildings are means of transferring forces and moments.



A building has two basic parts

- Sub structure or foundations
- Super-structure

Concepts of Foundation

A foundation is that part of the structure which is in direct contact with the ground to which the loads are transmitted. Super- structure is that part of the structure which is above ground level.

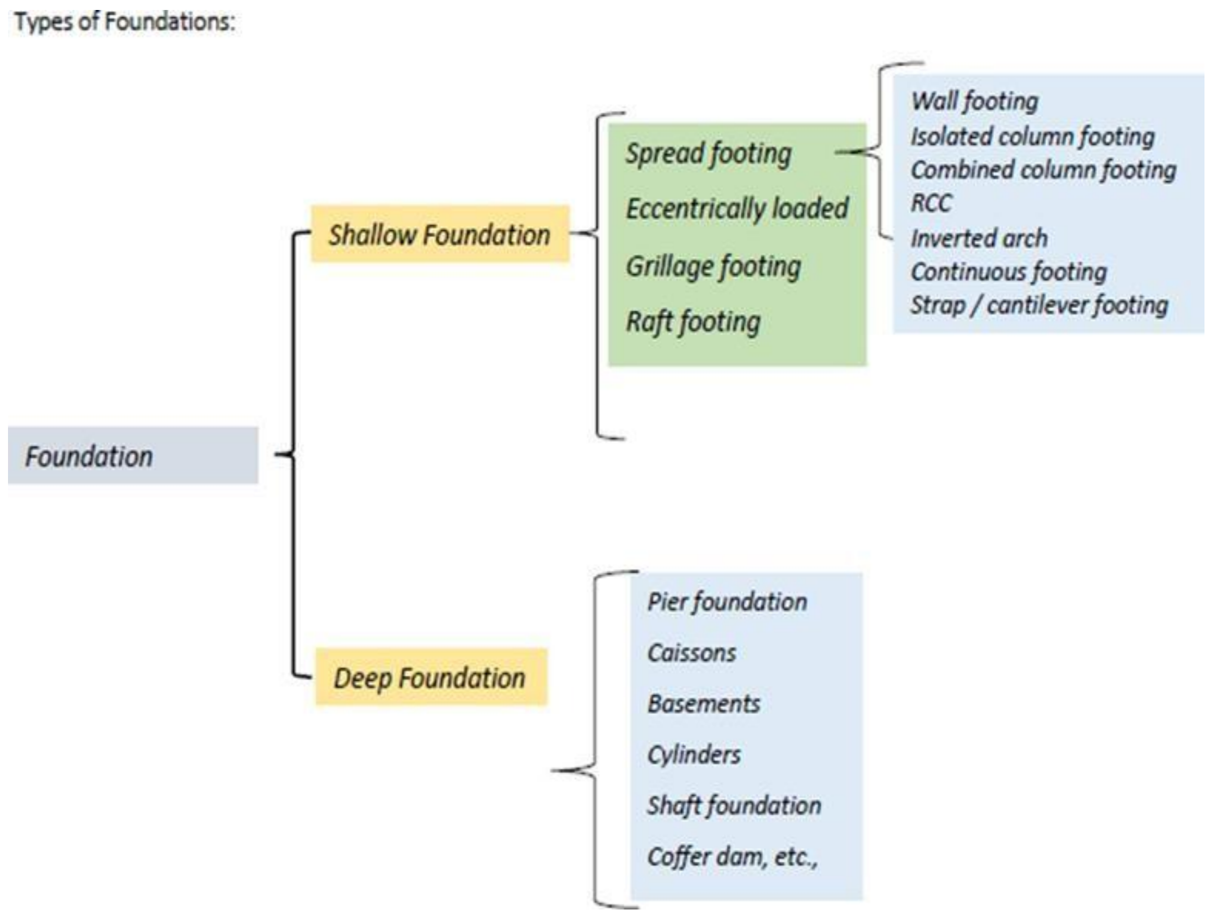
Functions of Foundation:

- **Reduction of load intensity:** Foundations distribute the loads of the Super-structure to a **larger area** So that the intensity of the load at its base **does not exceed** the safe bearing capacity of the Sub-soil.
- **Even distribution of load:** Foundations distribute the **non-uniform** load of the Super-structure evenly to sub-soil.
- **Provision of level surface:** It provides **leveled and hard** surface over which the super-structure can be built.
- **Lateral stability:** It anchors the super-structure to the ground, thus imparting lateral stability (side to side forces) to the super-structure.
- **Safety against undermining:** It provides the structural safety against undermining or scouring due to burrowing animals and flood water.
- **Protection against soil movements:** Special foundation measures prevent or minimize the **distress or cracks** in the super-structure due to expansion or contraction of the sub-soil.

Requirements of good Foundation

- The foundation shall be constructed **to sustain** the dead and imposed loads and to transmit these to the sub-soil.
- Foundation must be rigid so that **differential settlements** are minimized.
- Foundation should be taken **sufficiently deep** to guard the building against damage caused by swelling/ shrinking of sub-soil.
- Foundation should be **so located** that its performance may **not be affected** due to any unexpected future influence.

Types of foundations



Foundation may be broadly classified under two headings:

I) Shallow foundations or Open Foundation : Depth is equal to or less than its width
II) Deep foundations: Depth is greater than its width

SHALLOW or OPEN FOUNDATIONS

Shallow foundations are used when the **load of the structure is light** and the bearing capacity of soil **near the surface** is sufficient to support the load.

The depth of a shallow foundation is usually less than its width (generally up to 3 meters).

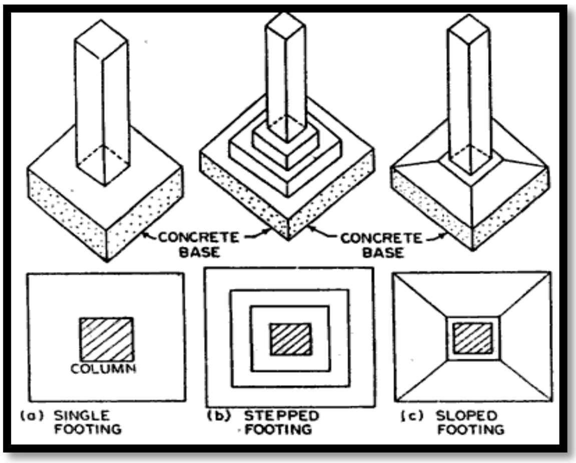
Shallow foundation may be of the following types:

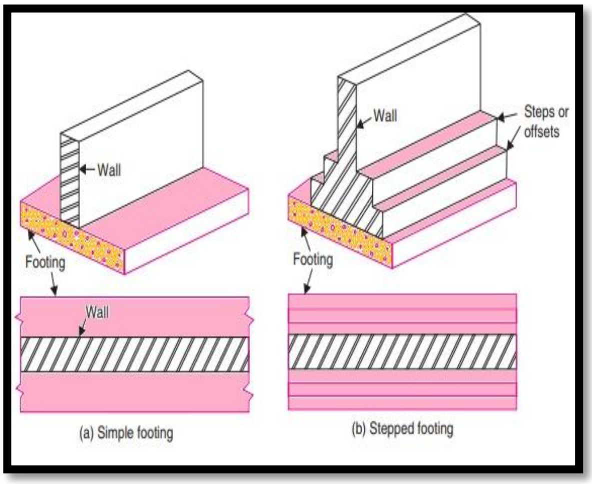
1. Spread Footing:

Spread footings are those which spread the super imposed load of wall or column over a larger area. It supports either individual columns or walls.

The various types of spread footings are:

- Single footing for a column
- Stepped footing for a column
- Sloped footing for a column
- Wall footing without step
- Stepped footing for wall
- Grillage foundation

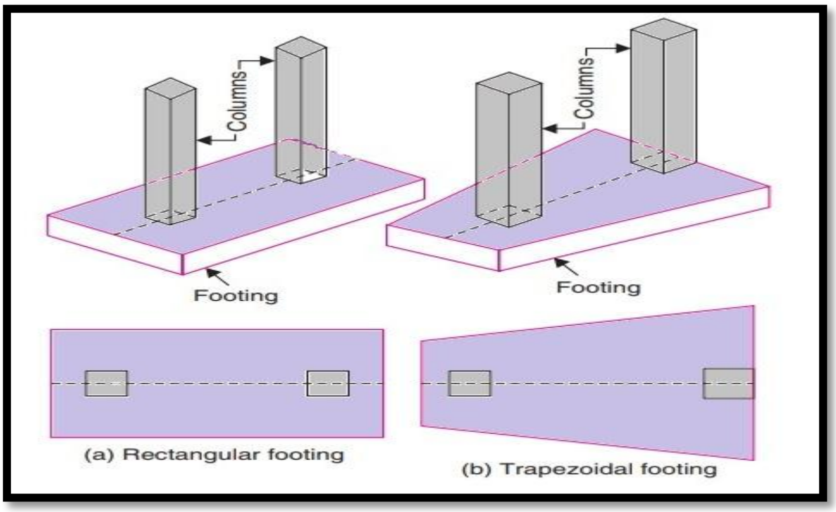




2. Combined Footing:

- Supports two or more columns that are close to each other.
- Used when columns are near the property line.

A combined footing supports two or more columns in a row. The combined footing can be rectangular in shape if both the columns **carry equal loads** or can be trapezoidal if both the **loads are unequal.** Generally they are constructed of reinforced concrete. The location of the center of the gravity of the column loads and centroid of the footing should coincide.



3. Strap (Cantilever) Footing:

- Two isolated footings are connected by a strap beam.
- Distributes load evenly when columns are close to property lines.

Strap footing consists of two or more individual footings connected by a beam called strap. This type of footing is used where the distance between the columns is so great that the **trapezoidal footing becomes quite narrow** with bending moments.

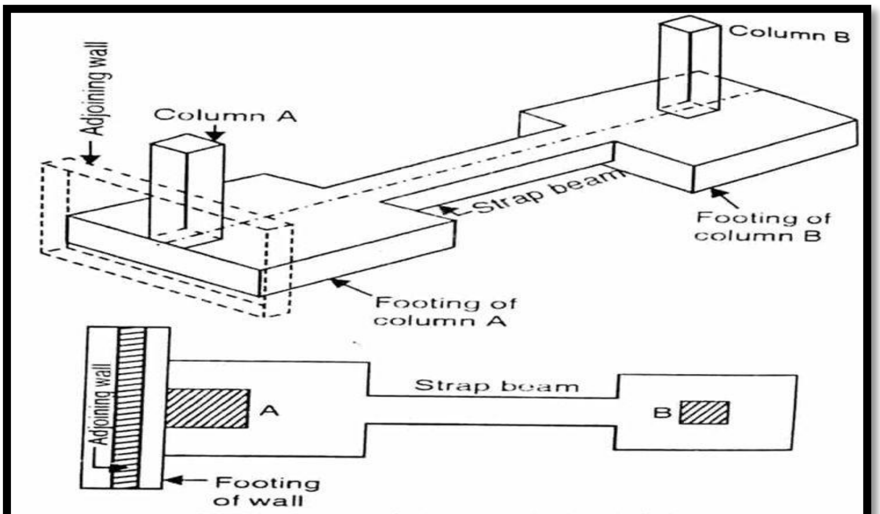
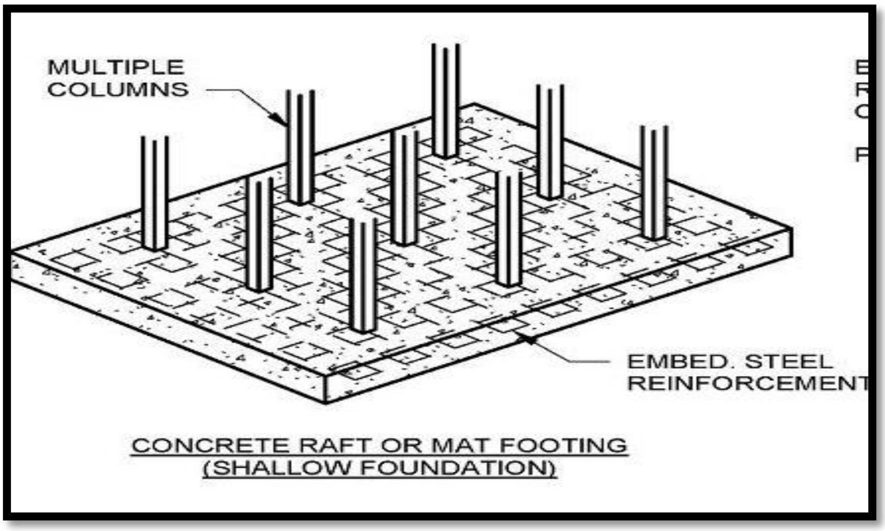


FIG. 2.7. STRAP FOOTING.

4. Raft or Mat Foundation:

- A thick reinforced concrete slab covering the entire building area.
- Used when loads are heavy or soil bearing capacity is low



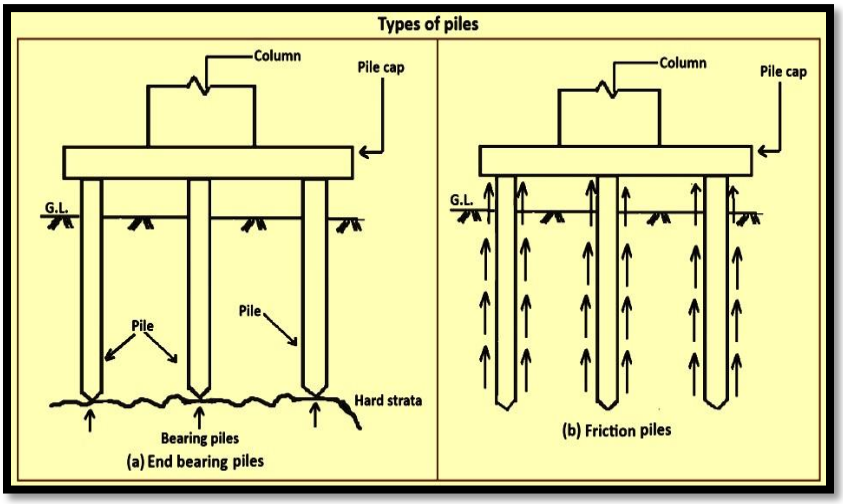
DEEP FOUNDATIONS

Deep foundations are used when the soil near the surface is **weak** or **compressible**, and loads must be transferred to **deeper, stronger soil layers or rock**. The **depth** of a deep foundation is **greater than its width** (more than 3 meters).

These foundations carry loads from a structure through **weak incompressible soils or fills** on to the **stronger and less compressible soils or rocks** at depth. These foundations are in general used as basements, buoyancy rafts, caissons, cylinders, shaft and piles.

1. Pile Foundation

A pile foundation consists of **long, slender columns** made of timber, steel, or reinforced concrete that are driven, bored, or cast into the ground to transfer structural loads to a stronger soil layer or rock at greater depth. Piles are used when the top soil has **low bearing capacity** or when the loads from the structure are **very heavy**. The load transfer mechanism of piles can be through end bearing, where the pile tip rests on a hard stratum, or through skin friction, where the load is transferred by frictional resistance along the surface of the pile. In some cases, piles work through both mechanisms. Piles can be precast and driven into the ground or cast in-situ using bored holes. They are commonly used for bridge foundations, marine structures, and high-rise buildings.



2. Pier Foundation

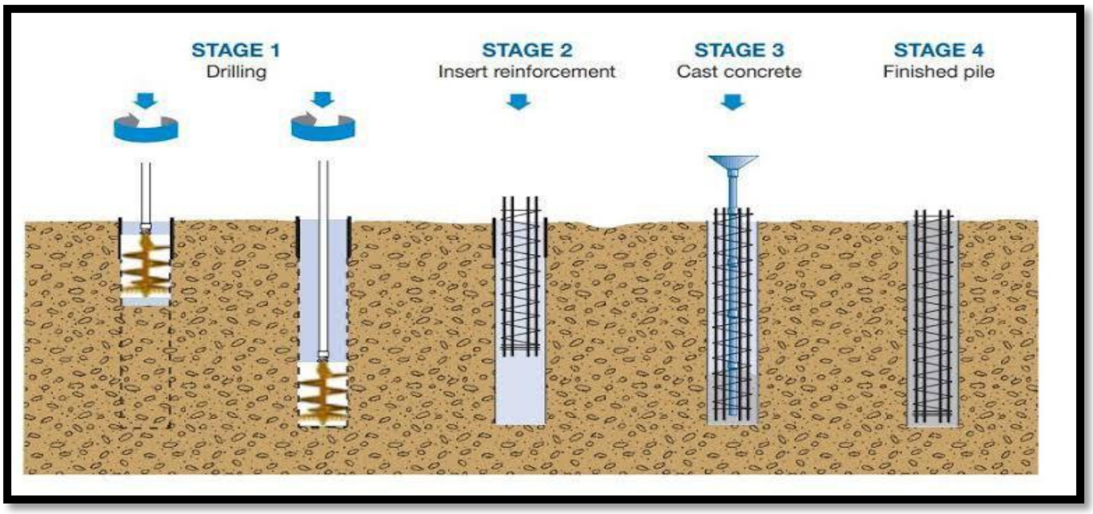
A pier foundation consists of **large-diameter cylindrical columns** made of reinforced concrete that transfer heavy loads to firm soil or rock at moderate depths. Unlike piles, piers are **shorter, have a larger diameter**, and are usually constructed in dry conditions by excavating **a hole** and filling it with **concrete.** Piers are often used where the surface soil is weak but a strong bearing layer is found within a reasonable depth. This type of foundation is suitable for bridges, multi-storey buildings, and industrial structures. Piers provide strong and stable support and are capable of carrying **heavy concentrated loads.**

3. Caissons or Well Foundation

A caisson, also known as a well foundation, is a large, **hollow**, cylindrical or rectangular structure made of reinforced concrete that is sunk into the ground or underwater bed until it reaches a stable stratum. It is commonly used in underwater construction such as bridge piers, abutments, and harbor works. The **caisson is first constructed on the ground surface** and then **gradually sunk** to the desired depth by **removing soil** from inside the well. Once it reaches the required level, it is **filled with concrete** to form a solid base. This type of foundation provides excellent stability and resistance against water currents, making it ideal for marine and river structure.

4. Drilled Shaft or Bored Pile Foundation

A drilled shaft foundation, also called a bored pile foundation, is formed by drilling a **deep cylindrical hole** into the ground and filling it with **reinforced concrete.** This method is used when **very large loads** need to be supported or when construction must be done with minimal noise and vibration, such as in urban areas. The load is transferred through **both end bearing and skin friction.** They are commonly used in the construction of high-rise buildings, transmission towers, and bridges



Causes of failure of good foundation:

1. **Non-uniform settlement** of subsoil and masonry
2. Alternative **swelling and shrinkage** of subsoil.
3. **Action of weathering agencies** like sun, rain, wind, and earthquake forces etc.
4. **Root trees and shrubs** which penetrate into the soil which has more affinity towards water which may leads to the failure of the foundation.

PLINTH

The **plinth** is the **portion of a building structure between the ground level and the floor level** immediately above the foundation. It acts as a **bridge between the substructure (foundation)** and the **superstructure (walls, columns, floors, etc.)** of the building. The plinth is usually made of **reinforced cement concrete (RCC)** or **brick masonry** and is built on top of the foundation to distribute the load of the superstructure evenly to the foundation below usually by about **300 mm to 600 mm**.

Functions of Plinth

- **Acts as a base for the superstructure –** It supports the walls and columns above it.
- **Distributes load –** It evenly spreads the load from the building to the foundation.
- **Prevents dampness –** By raising the floor above ground level, it stops moisture from seeping inside.
- **Protects from flooding and insects –** Prevents entry of rainwater, dust, and termites.
- **Improves appearance –** Gives the building a defined and stable base.

LINTEL

A lintel is a horizontal structural member **placed over openings** such as doors, windows, and ventilators in a building wall. Its main purpose is to support the **load of the wall** or structure above the opening and safely transfer it to the side walls or columns. Lintels **act as small beams** that span across the openings and prevent the **wall above from collapsing.**

- The **width** of lintel should be equal to the **width** of the **wall**.
- A proper bearing of lintel ends on supports is very essential. As a general rule, the bearing of the lintel at its ends should be either **10 cm** or **4.0 cm for every 30 cm** of span, whichever is greater.
- The depth of the lintel can be adopted as **1/12 th** of the span or **15 cm** whichever is greater.
- The lintels should be strong enough to resist failure due to the forces of **compression, tension and shear.**

CLASSIFICATION OF LINTELS

Lintels are classified into the following types, according to the materials of their construction:

1) **Wooden lintels**
2) **Stone lintels**

3) **Brick lintels**
4) **Reinforced concrete lintels**
5) **Steel lintels**

1. Wooden Lintels

A **wooden lintel** is the oldest and simplest form of lintel. It consists of a **single piece** of timber or **several wooden pieces** joined together to span an opening. These lintels are commonly used in small residential buildings and temporary structures. Although easy to construct and lightweight, wooden lintels are **not fire-resistant** and **can decay or get damaged by termites** over time. To prevent these issues, they are often treated with **preservative chemicals.** They are now rarely used due to the availability of stronger and more durable materials.

2. Stone Lintel

A **stone lintel** is made from a **single stone slab** placed over an opening. It is strong in compression and suitable for **stone masonry buildings**. Stone lintels are usually **thick and heavy**, and therefore used only where good quality stone is available locally. They are **not suitable for large spans** because stone has **low tensile strength** and can **crack under bending stress**. Stone lintels are often used in traditional or heritage constructions for their aesthetic appeal.

3. Brick Lintel

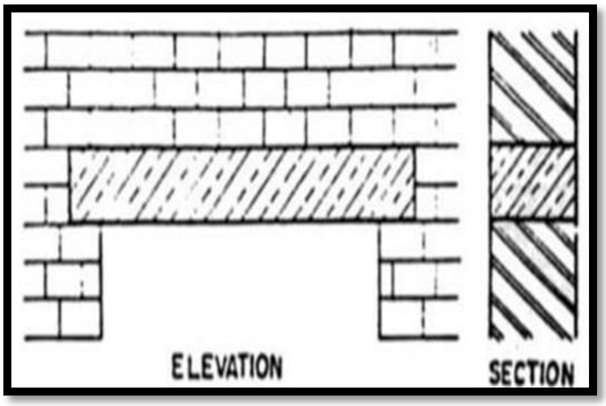
A **brick lintel** is constructed using hard, well-burnt bricks laid on edge or end with cement mortar. For small openings (less than 1 meter), plain brick lintels may be sufficient. However, for larger spans, **reinforced brick lintels** are used, where **steel bars** or **flat iron strips** are embedded between brick courses to increase strength. Brick lintels are economical, easy to construct, and match well with brick masonry walls, but they are **not suitable for heavy loads or wide openings**.

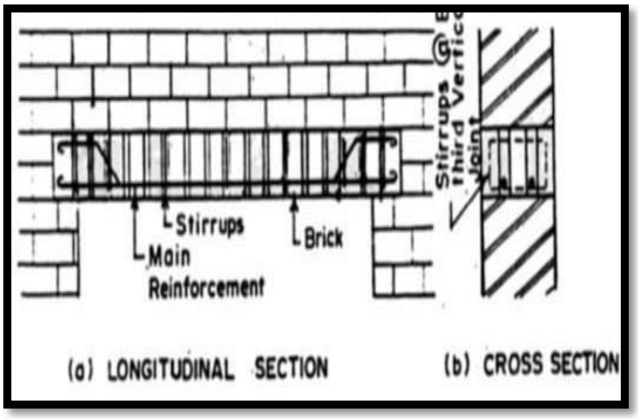
4. Reinforced Cement Concrete (RCC) Lintel

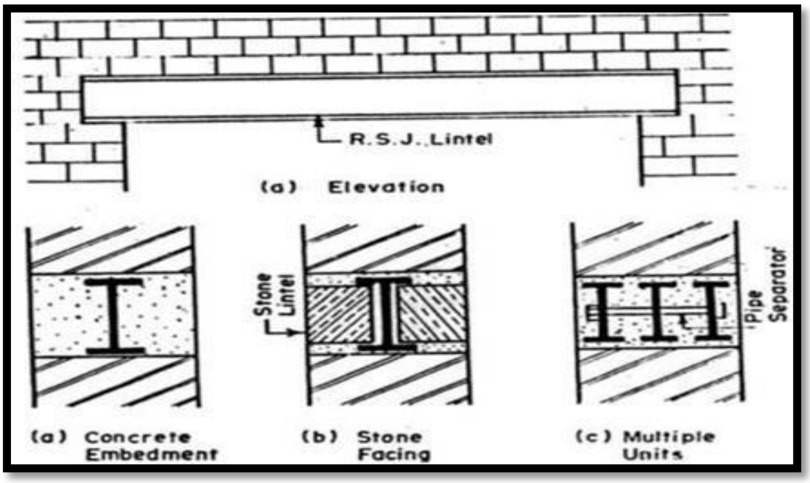
An **RCC lintel** is the **most commonly used type** in modern buildings. It is made of concrete reinforced with steel bars to withstand both compression and tension. RCC lintels can be **cast in place or precast** and are suitable for **long spans and heavy loads.** They are **fire-resistant, durable, and economical** in maintenance. RCC lintels can also be combined with sunshades or chajjas above windows. Because of their strength and adaptability, they are preferred in almost all types of modern structures.

5. Steel Lintel

A **steel lintel** consists of **rolled steel sections**, such as angles, channels, or I-sections, placed over the opening. Steel lintels are used where **large spans** or **heavy loads** need to be supported. They are light in weight and easy to install. Sometimes, a **steel plate** is embedded in concrete to form a **composite lintel** for additional strength and protection against corrosion. Steel lintels are ideal for industrial buildings and structures where strength and long spans are important.







CHEJJA

A **chejja**, also called a **chajja**, is a **projected slab or shading structure** built **above doors, windows, or along exterior walls** of a building. It is usually made of **reinforced cement concrete (RCC)** or sometimes **stone slabs** in traditional buildings. The chejja acts as a **protective covering** that project horizontally from the wall surface to **shield openings from rain, sunlight, and dust**.

Purpose of chejja:

- The main purpose of providing a chejja is to **protect the wall and openings** such as doors and windows from **direct rainwater, heat, and glare of sunlight**.
- It **prevents water** from entering the interior of the building and helps in maintaining a cool indoor temperature.
- Chejjas also **improve the durability** of the external plaster and paint by minimizing the impact of weathering.
- In addition, they contribute to the **aesthetic appearance** of the façade by giving a neat and decorative look to the building.

MASONRY

Masonry is the **art and craft of building structures** using individual units such as bricks, stones, concrete blocks, or other similar materials, which are **laid and bonded together with mortar.** It forms the main part of walls, columns, foundations, and arches in most buildings. Masonry construction has been used since ancient times because it provides strength, durability, fire resistance, and thermal insulation to structures.

Depending on the type of building units used, masonry may be classified into the following:

- Stone masonry
- Brick masonry
- Reinforced brick masonry
- Hallow concrete blocks masonry
- Composite masonry

Brick masonry

Constructed using **bricks and mortar,** this is the most common type of masonry used in buildings. It is economical, easy to handle, and gives a uniform appearance. Brick masonry can be **load-bearing or non-load-**

bearing, depending on the design.

General principles in brick masonry construction:

- **Good brick masonry** should utilize bricks, which are sound, hard, well burnt and tough with uniform colour, shape and size.
- The bricks should be compact, homogeneous, free form holes, cracks, flaws, air-bubbles and stone lumps and **soaked in water** for at least two hours before use
- In the brickwork, the bricks should be laid on their beds with the **frogs pointing upwards.**
- The brick courses should be **laid truly horizontal** and should have truly vertical joints as far as possible the use of **brick – bats** should be discouraged as far as possible, the brick wall should be **raised uniformly less than 1.5m** in day with proper bond.
- When the mortar is green the face joints should be racked to a **depth of 12 to 19mm** in order to have a proper key for **plastering or pointing.**
- In order to ensure continuous bond between the old and the new, the wall should be stopped **with a toothed end.**

Masonry Wall Requirements

The usual functional requirements of a masonry wall include:

- Adequate strength to **support imposed loads**
- Sufficient **water tightness**
- Sufficient **visual privacy** and **sound** transmission
- Appropriate **fire resistance**
- Ability to **accommodate** heating, air conditioning, electrical, and plumbing equipment
- Ability to receive various finish materials Cost
- Ability to **provide openings** such as doors and window.

Characteristics of brick bond or rules for bonding:

- The brick masonry should have bricks of uniform shape and size
- For satisfactory bondage the lap should be one-fourth of the brick along the length of the wall and half brick across thickness of the wall
- The brick bats use should be discouraged
- The vertical joints in the alternate courses should coincides with the centre line of the stretcher
- The alternate courses the centre line of header should coincide with the centre line of stretcher, in course below or above it.
- The stretcher should be used only in the facing while hearting should be done in the headers only

Classifications of bonds:

The bonds can be classified as follows:

- Stretcher bond
- Header bond
- English bond
- Double Flemish bond
- Single Flemish bond

COLUMN

A column or pillar is a vertical compression member used in buildings and other structures to support loads from beams, slabs, or arches and transfer them safely to the foundation. Columns are essential elements in any framed structure as they carry the axial loads and sometimes bending moments, ensuring the overall stability of the structure

Function of Column

The main functions of a column are:

- To transfer the load from beams, slabs, or roof structures down to the foundation.
- To provide vertical support and maintain the structural integrity of buildings.
- To resist compressive and buckling forces due to vertical loads.
- To keep the building components in proper alignment and position.
- In some cases, columns also serve decorative purposes in architecture (e.g., classical or ornamental columns).

BEAMS

A beam is a horizontal or inclined structural member that primarily carries transverse loads (loads acting perpendicular to its length). It transfers the load from the slab or floor above to the columns or walls below. Beams are subjected mainly to bending moments and shear forces during service.

Function of Beam

The main functions of a beam are:

- To support loads from slabs, floors, or roofs and transfer them to vertical supports (columns or walls).
- To resist bending moments developed due to transverse loading.
- To resist shear forces within the member.
- To provide stability and maintain the integrity of the structure.
- To distribute loads uniformly to supporting members.

FLOORING

Flooring is the bottom surface of a room or building on which occupants walk. It forms an important part of a structure, providing a level, strong, durable, and aesthetic surface that can safely carry the imposed loads of occupants, furniture, and equipment. A well-designed floor enhances comfort, appearance, cleanliness, and durability of a building.

Functions of Flooring

- Provide a hard and even surface for movement and activities.
- Distribute loads safely to the subfloor or foundation below.
- Resist wear and tear due to abrasion and impact.
- Provide an attractive appearance to interiors.
- Offer insulation against heat, cold, and noise.
- Prevent dampness from the ground.
- Ensure easy cleaning and maintenance.

Requirements of a Good Floor

- Durable and strong to withstand loads and traffic.
- Hard and wear-resistant to avoid abrasion.
- Smooth but not slippery for safety.
- Damp-proof to prevent moisture from rising.
- Easy to clean and maintain.
- Aesthetically pleasing in color and texture.
- Economical in cost and maintenance.

Components of Flooring

- **Subgrade:** The natural ground prepared to support the floor.
- **Base Course (Sub-base):** A layer of sand, gravel, or concrete over subgrade for load distribution.
- **Damp-Proof Course (DPC):** Prevents moisture from rising up.
- **Flooring Material (Top Layer):** The visible finished surface such as cement, tile, or wood.

SLABS

A concrete slab is a structural feature, usually of constant thickness, that can be used as a floor or a roof. A slab-on-ground is supported on the subsoil and is usually reinforced with reinforcing bars or welded wire mesh. A suspended slab (or structural slab) spans between supports and must be reinforced to resist bending moments calculated from statics based on the magnitude of load and span. There are one-way slabs, two- way slabs, waffle slabs, flat plates, flat slabs, and many other slab types.

Raw Materials for RCC Slab Roof

- Cement
- Coarse aggregate
- Fine aggregate
- M.S. Steel bar
- Binding wire
- Water
- Shuttering materials such as wooden Ballies, Planks, and Iron Plates etc.

Advantages of RCC Slab

- Energy efficient.
- Does not catch fire.
- Provides solid and durable roofing.
- Very versatile and provides greater protection.
- Reduces costs of insurance and has resale value.

STAIRS

The means of communication between various floors is offered by various structures such as stairs, lifts, ramps, ladders, escalators.

STAIR: A stair is a series of steps arranged in a manner as to connect different floors of a building. Stairs are designed to provide an easy and quick access to different floors.

- A staircase is an enclosure which contains the complete stairway.
- In a residential house stairs may be provided near the entrance.
- In a public building, stairs must be from main entrance and located centrally.

STAIRCASE: Room of a building where stair is located. STAIRWAY: Space occupied by the stair.

TECHNICAL TERMS

BALUSTER: Vertical member which is fixed between stairway and horizontal to provide support to hand rail.

BALUSTRADE: Combined framework of baluster and hand rail.

STRING: Inclined member of a stair which supports ends of steps. They are of two types,

(i) cut/open string,

(ii)(ii) closed/housed string.

- In open string, upper edge is cut away to receive the ends of steps.
- In closed string, the ends of steps are layed between straight and parallel edges of the string.

FLIGHT : Unbroken series of steps between the landings.

GOING: horizontal distance between faces of two consecutive risers.

HANDRAIL: inclined rail over the string. Generally it is moulded. It serves as a guard rail. It is provided at a convenient height so as to give grasp to hand during ascent and descent.

HEAD ROOM: vertical distance between nosings of one flight and the bottom of flight immediately above is called head room.

LANDING: horizontal platform between two flights of a stair. A landing facilitates change of direction and provides an opportunity to take rest.

NEWEL POST: vertical member placed at ends of flights to connect ends of string and hand rail.

NOSING: projection part of tread beyond face of riser.

LINE OF NOSING: imaginary line parallel to strings and tangential to nosings. The underface of hand rail should coincide with line of nosing.

PITCH: angle of inclination of stair with floor. Angle of inclination of line of nosing with horizontal.

RISE: vertical distance between two successive treads.

RISER: vertical member of the step, which is connected to treads.

RUN: length of a stair in a horizontal plane which includes length of landing.

SCOTIA: an additional finish provided to nosing to improve the elevation of the step which also provides strength to nosing.

SOFFIT: under surface of a stair. Generally it is covered with ceiling or finished with plaster.

STEP: combination of tread and riser. Tread: horizontal upper portion of a step.

Waist: thickness of structural slab in RCC stair

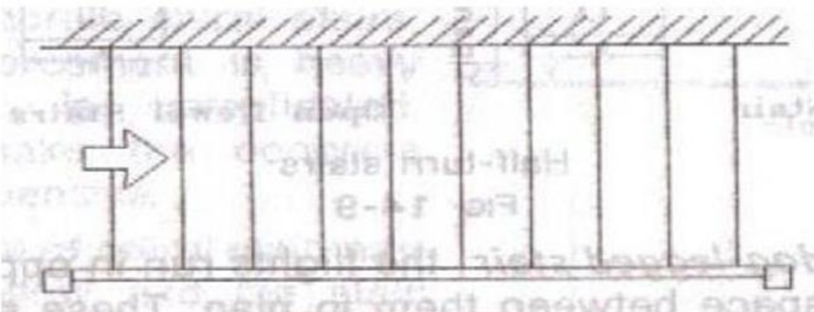
Carriage: a rough timber supporting steps of wooden stairs

Requirements of a good stair

1) **Location:** It should be so located as to provide easy access to the occupants of the building. It should be so located that it is well lighted and ventilated directly from exterior
2) **Width of stair:** It should be wide enough to carry the user without much crowd and inconvenience. Width of the stair depends on its location and the type of building. In a domestic building, a 90 cm wide stair is sufficient and for public building 1.4m to 1.8m may be required
3) **Length of flight:** For the comfortable ascent of stairway the number of steps in a flight should be restricted to a maximum of 12 and a minimum of 3.
4) **Pitch of stair:** In general the pitch of stair should never exceed 400 and should not be flatter than 250
5) **Head room:** The clear distance between the tread and the soffit of the flight immediately above it should be a minimum of 2.14m.
6) **Balustrade:** The open well stairs should be provided with balustrade so as to minimize the damage or accidents
7) **Step dimensions:** The rise and tread of each step in a stair should be of uniform dimension throughout so as to provide comfort to users (Going should be between 25-30cm and rise should be between 10- 15cm). The width of the landing should not be less than the width of stair
8) **Materials of construction:** The stair should preferably constructed of materials which posses fire-resisting qualities.

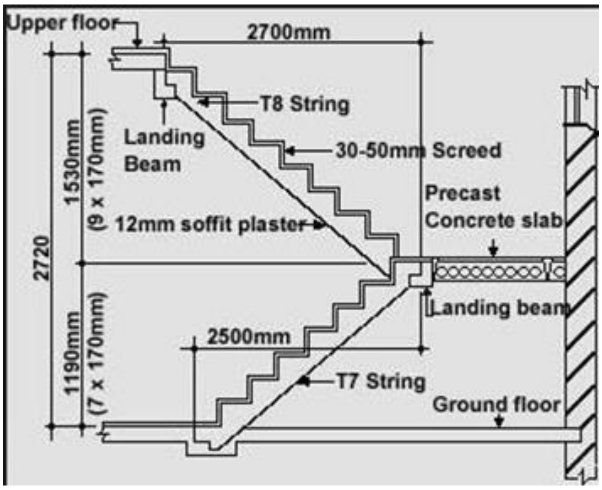
Classification of stairs

1. **Straight stairs:** In this form of stair, all steps rise in same direction. If the ascending is steep, the straight flight may be broken at an intermediate landing. This type stair is usefull where the stair case is long and narrow and the possibility of any other form of stair may not be practically possible.


Straight stair

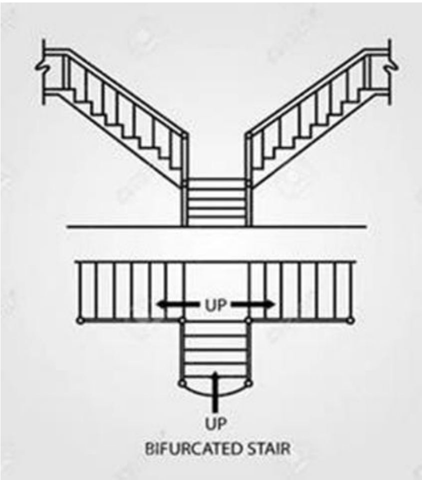
2. **Dog-legged stair:** The flights run in opposite direction and there is no space between them in plan. These stairs are useful where total width of space available for the staircase is equal to twice the width of space.

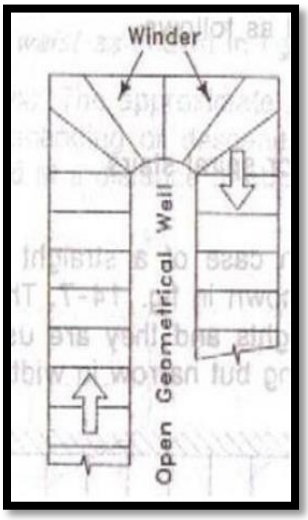
3. **Open-Newel stairs:** It consists of two or more straight flights arranged in such a manner that a clear space called a “well” occurs between the backward and forward flights.



4. **Bifurcated stairs:** In this type of stairs, the flights are so arranged that there is a wide flight at the start which is divided into narrow flights at the mid-landing. The two narrow flights start from either side of the mid-building.

5. **Circular or helical or spiral stairs:** In this type of stairs. The steps radiate from the centre and they do not have either any landing or intermediate newel post. This type of stairs is commonly provided at the backside of the building.





6. **Geometrical stairs:** This is similar to the open-newel stair with the difference that open well between the forward and backward flight is curved. In this form of stairs, change of direction is obtained through winders.